

EDA Report

Generated: 2026-04-08 17:35:48

Dataset Overview

- Data path: bank_aml_tm_synthetic_10k_local=data/bank_aml_tm_synthetic_10k_local.csv
- Rows used: 10000
- Target column: sar_label
- Time column: value_date

Data Quality

- Dataset has 10000 rows and 35 columns.
- Duplicate row ratio: 0.00%.
- Highest missing rate is 1.97% in 'merchant_category'.

Missingness

Column	Missing Count	Missing Rate
merchant_category	197	1.97%
counterparty_account_id	160	1.60%
device_type	115	1.15%

The following columns have no missing values: txn_id, txn_datetime, customer_id, account_id, txn_amount, currency, origin_country, destination_country, txn_channel, payment_rail (+22 more) (32 columns total).

Null-like / Placeholder Values

These values are not nulls but may represent missing or invalid data placeholders.

Column	Null-like Count	Null-like Rate	Example Values
merchant_category	57	0.57%	unknown
device_type	55	0.55%	unknown

Column Type Classification

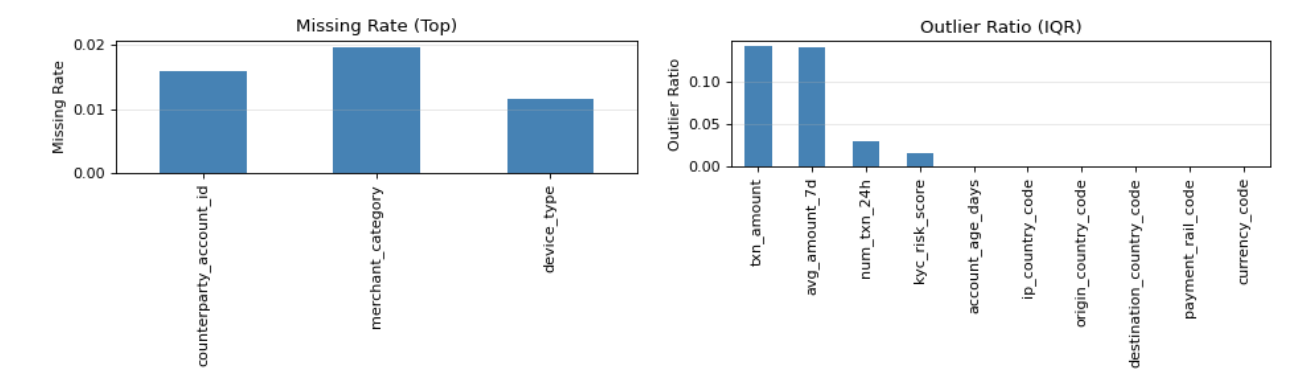
Type	Columns
numeric	txn_amount
numeric	account_age_days
numeric	kyc_risk_score
numeric	num_txn_24h
numeric	avg_amount_7d
numeric	origin_country_code
numeric	destination_country_code
numeric	ip_country_code
numeric	currency_code
numeric	txn_channel_code
numeric	payment_rail_code
numeric	txn_type_code

Type	Columns
numeric	merchant_category_code
numeric	device_type_code
categorical	currency
categorical	origin_country
categorical	destination_country
categorical	txn_channel
categorical	payment_rail
categorical	txn_type
categorical	merchant_category
categorical	device_type
categorical	ip_country
datetime	txn_datetime
datetime	value_date
boolean	is_pep
boolean	sanctions_match
boolean	is_cross_border
boolean	is_fintech_rail
boolean	sar_label

Outlier Ratio (IQR)

Column	Outlier Ratio
txn_amount	0.141
avg_amount_7d	0.140
num_txn_24h	0.030
kyc_risk_score	0.017
account_age_days	0
ip_country_code	0
origin_country_code	0
destination_country_code	0
payment_rail_code	0
currency_code	0

Charts



Target

- Target 'sar_label' has 2 classes.
- Target rate over time plotted.

Target Distribution

Value	Count	Rate
0	8,426	84.26%
1	1,574	15.74%

Target Rate by origin_country

origin_country	Target Rate
NG	30.81%
MX	24.64%
MA	21.24%
AE	19.75%
TR	19.50%
ES	16.49%
SG	16.26%
HK	15.56%
CN	15.46%
US	14.90%

Target Rate by destination_country

destination_country	Target Rate
NG	36.83%
MA	35.45%
AE	32.16%
MX	32.14%
TR	31.70%
CH	11.59%
HK	10.26%
BR	7.88%
GB	7.84%
CN	7.19%

Target Rate by ip_country

ip_country	Target Rate
IN	19.74%
CN	19.41%
MX	18.73%
HK	18.59%
DE	17.79%
BR	17.78%
IT	17.40%

ip_country	Target Rate
CH	16.38%
NG	15.33%
FR	15.20%

Target Rate by currency

currency	Target Rate
NGN	30.81%
MXN	24.64%
MAD	21.24%
AED	19.75%
TRY	19.50%
SGD	16.26%
HKD	15.56%
CNY	15.46%
USD	14.90%
INR	14.76%

Target Rate by payment_rail

payment_rail	Target Rate
swift	29.99%
wire	27.77%
zelle	26.79%
cashapp	23.66%
adyen	14.74%
paypal	13.16%
google_pay	12.66%
apple_pay	10.44%
venmo	8.80%
stripe	7.25%

Target Rate by merchant_category

merchant_category	Target Rate
travel	21.99%
luxury	20.85%
UNKNOWN	19.30%
utilities	17.45%
restaurants	15.22%
rideshare	15.09%
gaming	14.88%
grocery	13.79%
electronics	13.70%
crypto	12.82%

Target Rate by txn_type

txn_type	Target Rate
international_remit	36.17%
crypto_exchange	33.01%
cash_withdrawal	20.79%
p2p_transfer	9.22%
loan_payment	6.85%
merchant_pay	6.74%
cash_deposit	6.64%
bill_pay	6.61%

Target Rate by txn_channel

txn_channel	Target Rate
api	17.91%
call_center	16.36%
web	16.20%
mobile_app	15.51%
branch	12.85%

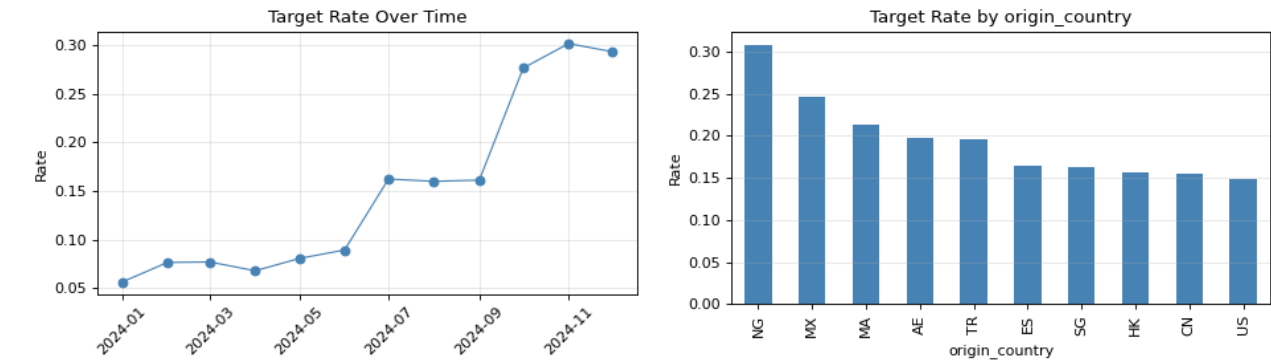
Target Rate by device_type

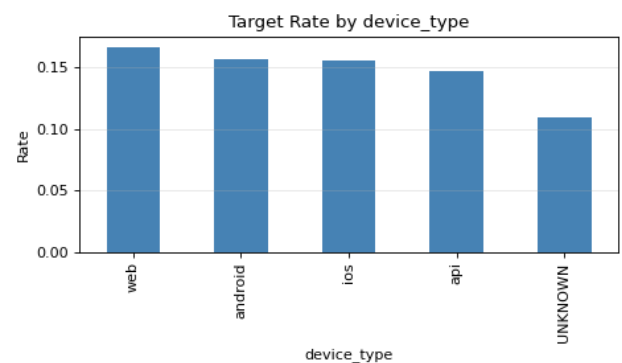
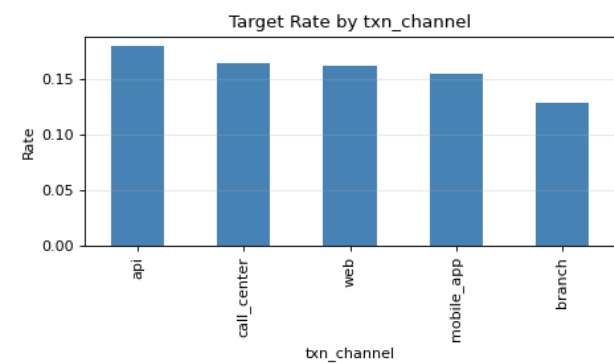
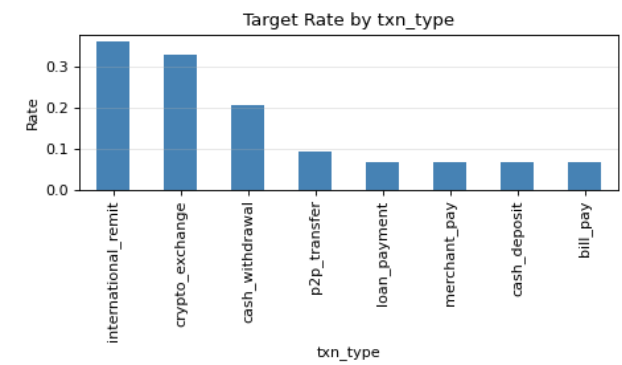
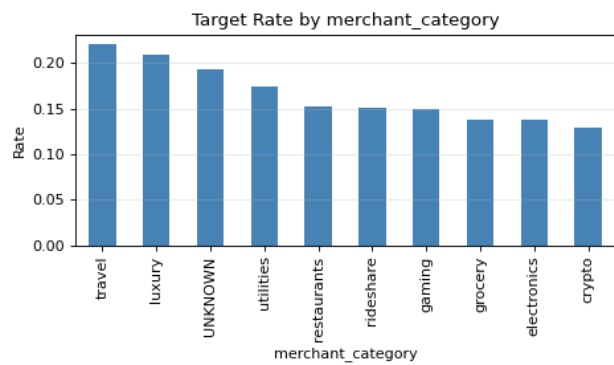
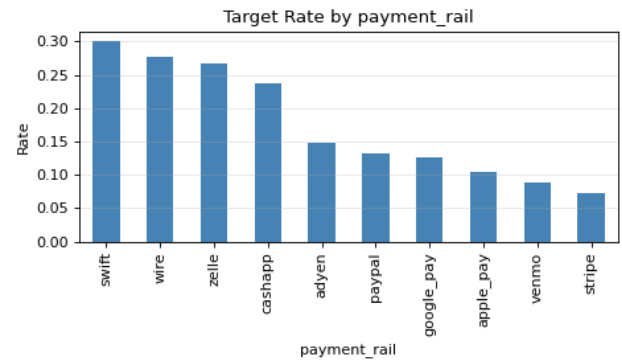
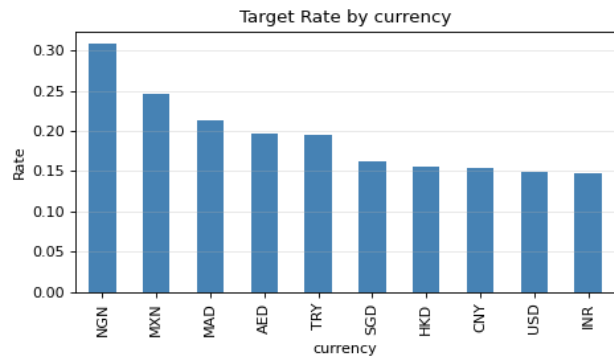
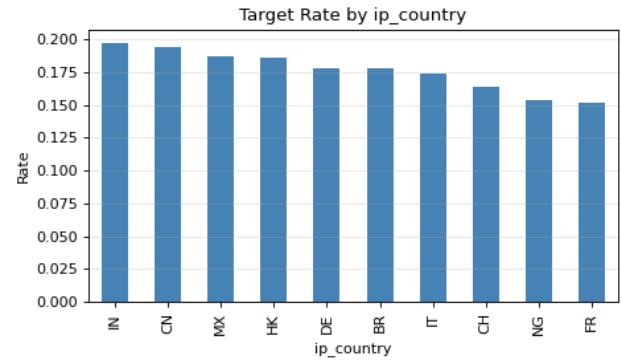
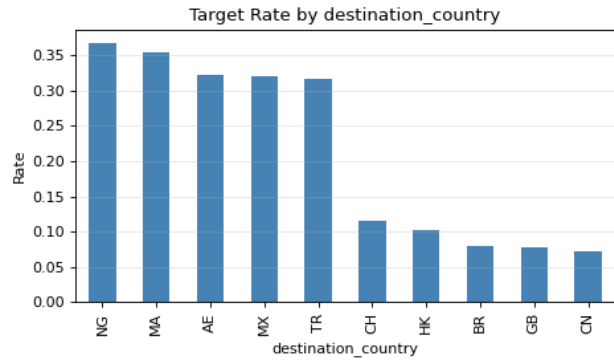
device_type	Target Rate
web	16.71%
android	15.67%
ios	15.57%
api	14.76%
UNKNOWN	10.91%

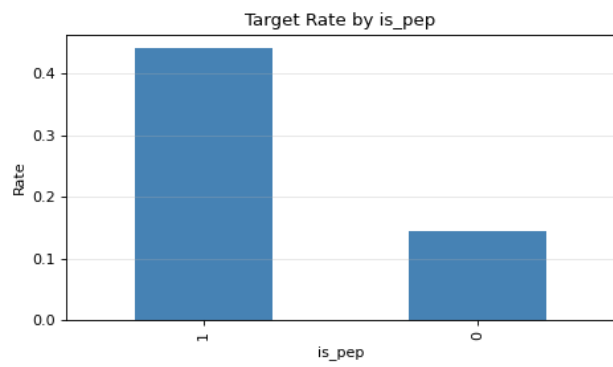
Target Rate by is_pep

is_pep	Target Rate
1	44.13%
0	14.48%

Charts







Univariate - Summary

Univariate

- Numeric columns analyzed: txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h, ip_country_code, origin_country_code, destination_country_code, payment_rail_code, currency_code
- Categorical columns analyzed: origin_country, destination_country, ip_country, currency, payment_rail, merchant_category, txn_type, txn_channel, device_type, is_pep

Numeric Summary Statistics

Column	count	mean	std	min	25%	50%	75%	max
txn_amount	10,000	1,005.141	7,490.498	1.550	45.392	126.210	416.035	410,040.160
avg_amount_7d	10,000	930.239	6,525.060	3.400	55.127	118.010	359.918	304,796.950
account_age_days	10,000	1,812.333	1,043.807	1	919	1,813	2,702	3,649
kyc_risk_score	10,000	40.010	18.243	0	27.619	38.852	50.841	100
num_txn_24h	10,000	4.330	8.082	0	1	3	5	85
ip_country_code	10,000	10.027	5.754	0	5	10	17	17
origin_country_code	10,000	10.040	5.727	0	5	10	17	17
destination_country_code	10,000	10.066	5.658	0	5	11	16	17
payment_rail_code	10,000	6.751	4.509	0	3	7	11	13
currency_code	10,000	7.241	4.342	0	4	5	13	13

Categorical Frequency (Top K)

Top K: origin_country

Category	Count	Rate
US	2,771	27.71%
DE	837	8.37%
GB	809	8.09%
FR	657	6.57%
IT	557	5.57%
ES	552	5.52%
IN	420	4.20%
NL	402	4.02%
CN	401	4.01%
BR	390	3.90%

Top K: destination_country

Category	Count	Rate
US	2,126	21.26%
MA	739	7.39%
AE	684	6.84%
TR	675	6.75%
MX	672	6.72%
GB	663	6.63%
DE	655	6.55%

Category	Count	Rate
NG	581	5.81%
FR	506	5.06%
IT	427	4.27%

Top K: ip_country

Category	Count	Rate
US	2,865	28.65%
DE	877	8.77%
GB	782	7.82%
FR	691	6.91%
ES	570	5.70%
IT	523	5.23%
NL	445	4.45%
CN	407	4.07%
BR	405	4.05%
IN	385	3.85%

Top K: currency

Category	Count	Rate
EUR	3,005	30.05%
USD	2,771	27.71%
GBP	809	8.09%
INR	420	4.20%
CNY	401	4.01%
BRL	390	3.90%
AED	324	3.24%
TRY	318	3.18%
MAD	306	3.06%
SGD	289	2.89%

Top K: payment_rail

Category	Count	Rate
wire	1,678	16.78%
ACH	1,666	16.66%
paypal	1,087	10.87%
swift	877	8.77%
card	841	8.41%
zelle	810	8.10%
cashapp	634	6.34%
rtp	543	5.43%
venmo	466	4.66%
sepa	422	4.22%

Top K: merchant_category

Category	Count	Rate
restaurants	1,813	18.13%
grocery	1,458	14.58%
utilities	1,370	13.70%
rideshare	1,173	11.73%
electronics	978	9.78%
gaming	800	8.00%
travel	764	7.64%
luxury	585	5.85%
charity	472	4.72%
crypto	390	3.90%

Top K: txn_type

Category	Count	Rate
p2p_transfer	2,376	23.76%
merchant_pay	1,647	16.47%
international_remit	1,493	14.93%
bill_pay	1,271	12.71%
crypto_exchange	1,018	10.18%
cash_withdrawal	967	9.67%
loan_payment	686	6.86%
cash_deposit	542	5.42%

Top K: txn_channel

Category	Count	Rate
mobile_app	4,506	45.06%
web	2,987	29.87%
api	1,016	10.16%
branch	996	9.96%
call_center	495	4.95%

Top K: device_type

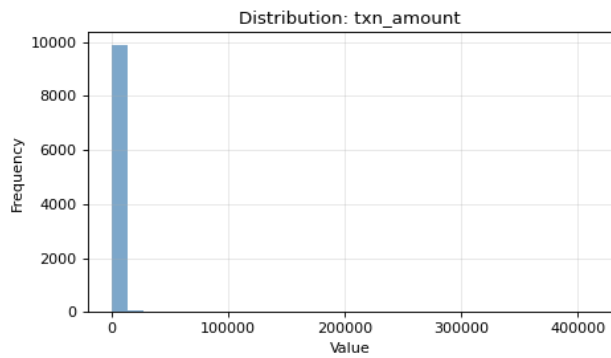
Category	Count	Rate
android	3,446	34.46%
ios	2,992	29.92%
web	2,424	24.24%
api	1,023	10.23%
nan	60	0.60%
UNKNOWN	55	0.55%

Top K: is_pep

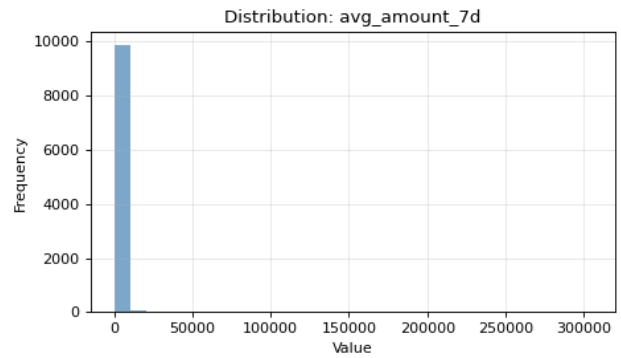
Category	Count	Rate
0	9,574	95.74%
1	426	4.26%

Univariate Plots

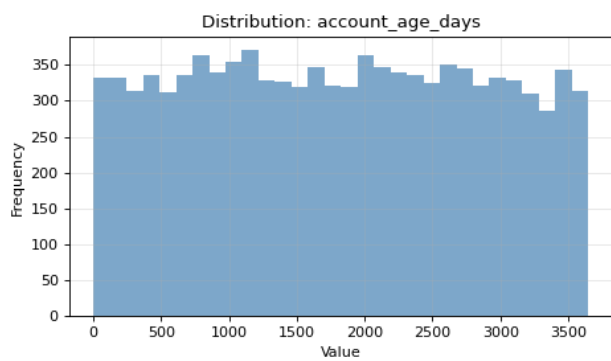
txn_amount distribution



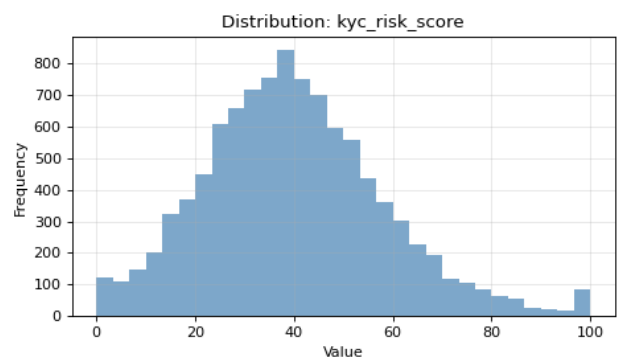
avg_amount_7d distribution



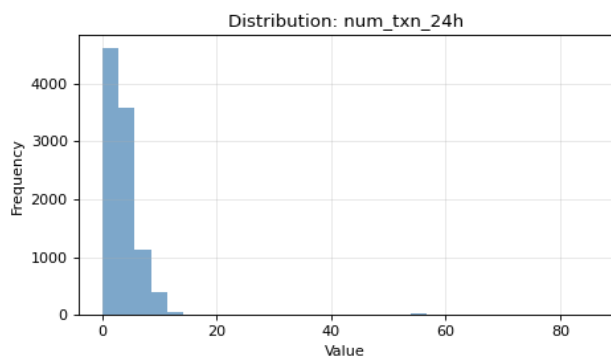
account_age_days distribution



kyc_risk_score distribution



num_txn_24h distribution



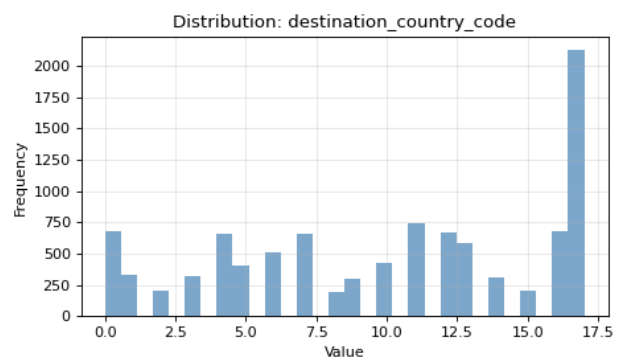
ip_country_code distribution



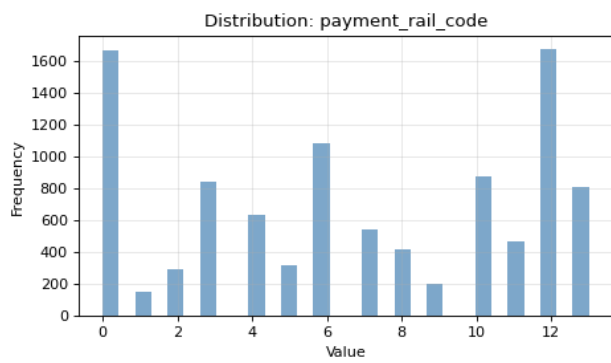
origin_country_code distribution



destination_country_code distribution



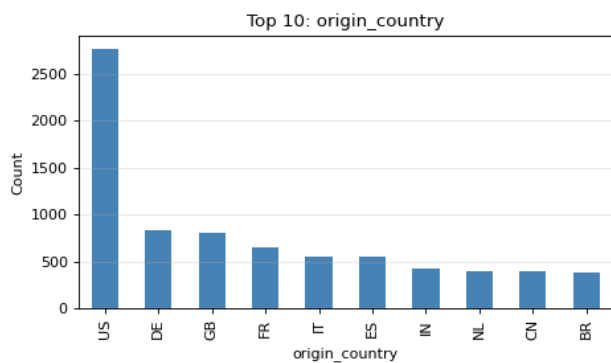
payment_rail_code distribution



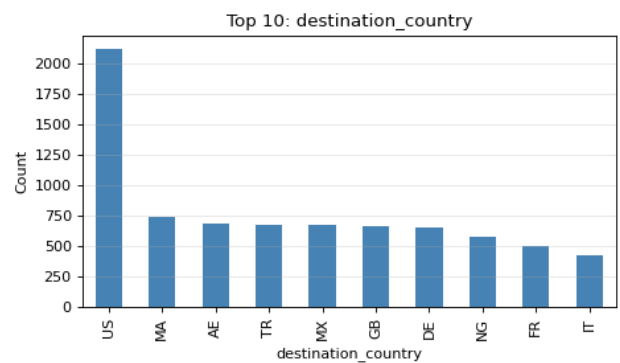
currency_code distribution



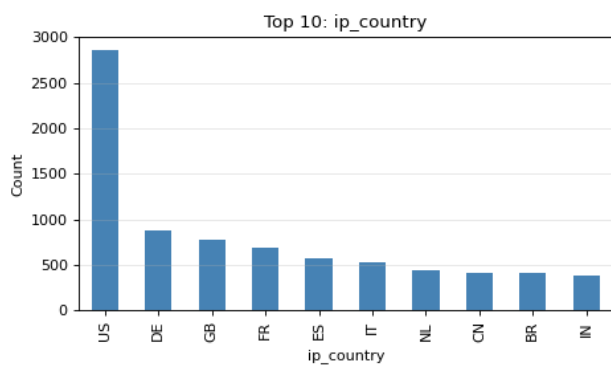
origin_country top 10



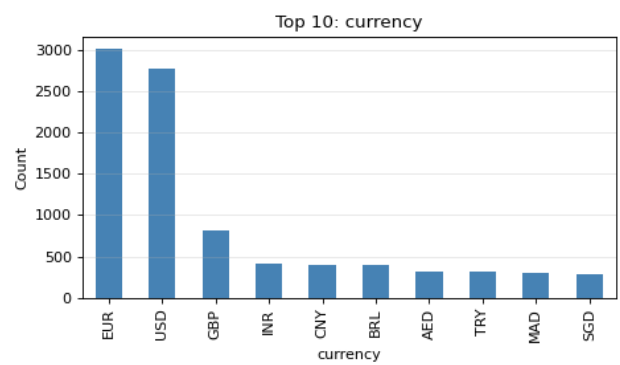
destination_country top 10



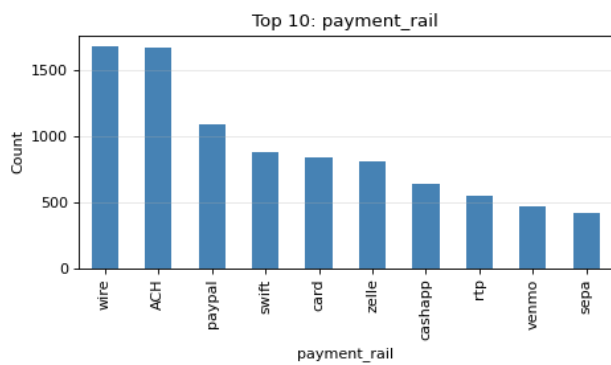
ip_country top 10



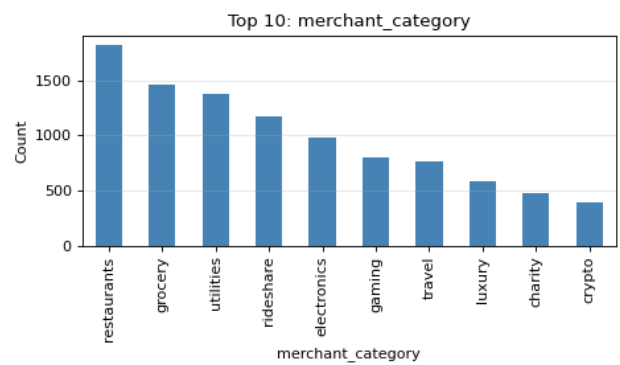
currency top 10



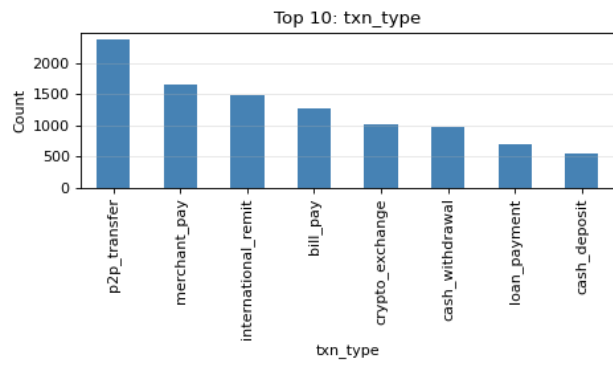
payment_rail top 10



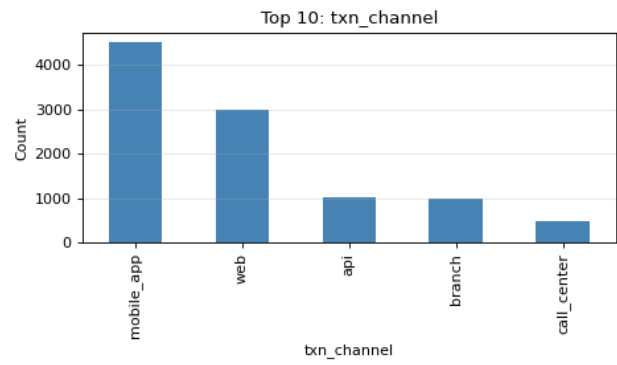
merchant_category top 10



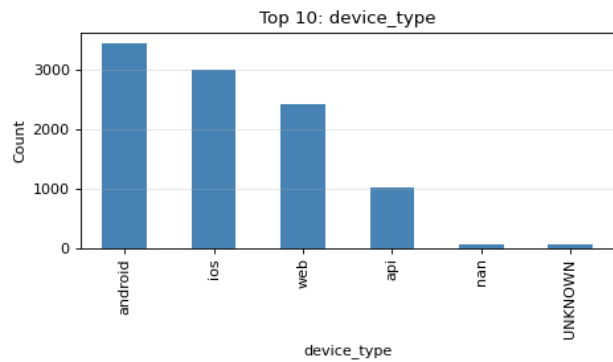
txn_type top 10



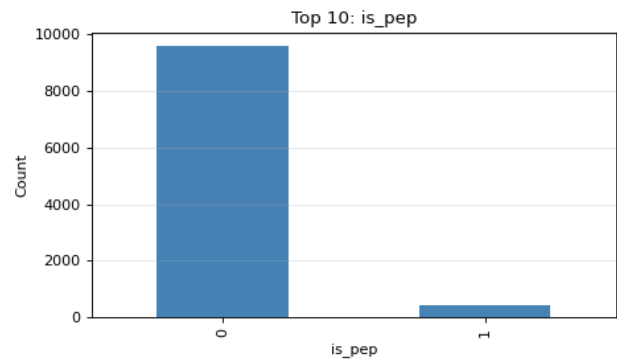
txn_channel top 10



device_type top 10



is_pep top 10



Bivariate Target

- Bivariate target analysis completed.

Numeric vs Target (Binned)

Column	Bin	Target Rate
txn_amount	(1.5490000000000002, 35.978]	4.40%
txn_amount	(35.978, 85.028]	8.60%
txn_amount	(85.028, 194.514]	12.15%
txn_amount	(194.514, 586.144]	19.80%
txn_amount	(586.144, 410040.16]	33.75%
avg_amount_7d	(3.399, 46.56]	6.20%
avg_amount_7d	(46.56, 85.432]	8.70%
avg_amount_7d	(85.432, 170.726]	12.45%
avg_amount_7d	(170.726, 495.312]	17.95%
avg_amount_7d	(495.312, 304796.95]	33.40%
account_age_days	(0.999, 740.0]	14.84%
account_age_days	(740.0, 1442.2]	15.66%
account_age_days	(1442.2, 2168.0]	17.12%
account_age_days	(2168.0, 2888.0]	15.11%
account_age_days	(2888.0, 3649.0]	15.97%
kyc_risk_score	(-0.001, 24.886]	5.70%
kyc_risk_score	(24.886, 34.63]	9.99%
kyc_risk_score	(34.63, 43.154]	11.96%
kyc_risk_score	(43.154, 54.013]	17.00%
kyc_risk_score	(54.013, 100.0]	34.05%
num_txn_24h	(-0.001, 1.0]	5.95%
num_txn_24h	(1.0, 2.0]	7.54%
num_txn_24h	(2.0, 3.0]	10.75%
num_txn_24h	(3.0, 5.0]	16.38%
num_txn_24h	(5.0, 85.0]	42.77%
ip_country_code	(-0.001, 4.0]	17.41%
ip_country_code	(4.0, 7.0]	14.64%
ip_country_code	(7.0, 12.0]	17.69%
ip_country_code	(12.0, 17.0]	14.53%
origin_country_code	(-0.001, 4.0]	14.86%
origin_country_code	(4.0, 7.0]	14.52%
origin_country_code	(7.0, 12.0]	17.43%
origin_country_code	(12.0, 17.0]	16.09%
destination_country_code	(-0.001, 4.0]	15.21%
destination_country_code	(4.0, 9.0]	7.59%
destination_country_code	(9.0, 12.0]	27.53%
destination_country_code	(12.0, 17.0]	14.80%
payment_rail_code	(-0.001, 2.0]	7.27%
payment_rail_code	(2.0, 6.0]	12.65%

Column	Bin	Target Rate
payment_rail_code	(6.0, 9.0]	5.89%
payment_rail_code	(9.0, 12.0]	25.49%
payment_rail_code	(12.0, 13.0]	26.79%
currency_code	(-0.001, 4.0]	14.72%
currency_code	(4.0, 8.0]	15.74%
currency_code	(8.0, 13.0]	16.90%

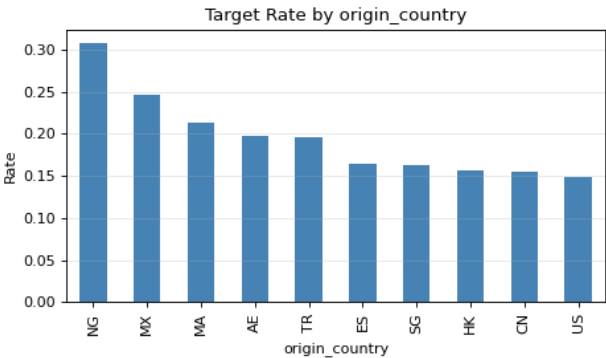
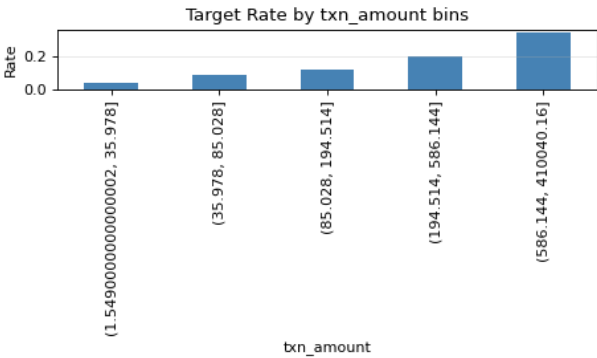
Categorical vs Target

Column	Category	Target Rate
origin_country	NG	30.81%
origin_country	MX	24.64%
origin_country	MA	21.24%
origin_country	AE	19.75%
origin_country	TR	19.50%
origin_country	ES	16.49%
origin_country	SG	16.26%
origin_country	HK	15.56%
origin_country	CN	15.46%
origin_country	US	14.90%
destination_country	NG	36.83%
destination_country	MA	35.45%
destination_country	AE	32.16%
destination_country	MX	32.14%
destination_country	TR	31.70%
destination_country	CH	11.59%
destination_country	HK	10.26%
destination_country	BR	7.88%
destination_country	GB	7.84%
destination_country	CN	7.19%
ip_country	IN	19.74%
ip_country	CN	19.41%
ip_country	MX	18.73%
ip_country	HK	18.59%
ip_country	DE	17.79%
ip_country	BR	17.78%
ip_country	IT	17.40%
ip_country	CH	16.38%
ip_country	NG	15.33%
ip_country	FR	15.20%
currency	NGN	30.81%
currency	MXN	24.64%
currency	MAD	21.24%
currency	AED	19.75%

Column	Category	Target Rate
currency	TRY	19.50%
currency	SGD	16.26%
currency	HKD	15.56%
currency	CNY	15.46%
currency	USD	14.90%
currency	INR	14.76%
payment_rail	swift	29.99%
payment_rail	wire	27.77%
payment_rail	zelle	26.79%
payment_rail	cashapp	23.66%
payment_rail	adyen	14.74%
payment_rail	paypal	13.16%
payment_rail	google_pay	12.66%
payment_rail	apple_pay	10.44%
payment_rail	venmo	8.80%
payment_rail	stripe	7.25%
merchant_category	travel	21.99%
merchant_category	luxury	20.85%
merchant_category	UNKNOWN	19.30%
merchant_category	utilities	17.45%
merchant_category	restaurants	15.22%
merchant_category	rideshare	15.09%
merchant_category	gaming	14.88%
merchant_category	grocery	13.79%
merchant_category	electronics	13.70%
merchant_category	crypto	12.82%
txn_type	international_remit	36.17%
txn_type	crypto_exchange	33.01%
txn_type	cash_withdrawal	20.79%
txn_type	p2p_transfer	9.22%
txn_type	loan_payment	6.85%
txn_type	merchant_pay	6.74%
txn_type	cash_deposit	6.64%
txn_type	bill_pay	6.61%
txn_channel	api	17.91%
txn_channel	call_center	16.36%
txn_channel	web	16.20%
txn_channel	mobile_app	15.51%
txn_channel	branch	12.85%
device_type	web	16.71%
device_type	android	15.67%
device_type	ios	15.57%
device_type	api	14.76%

Column	Category	Target Rate
device_type	UNKNOWN	10.91%
is_pep	1	44.13%
is_pep	0	14.48%

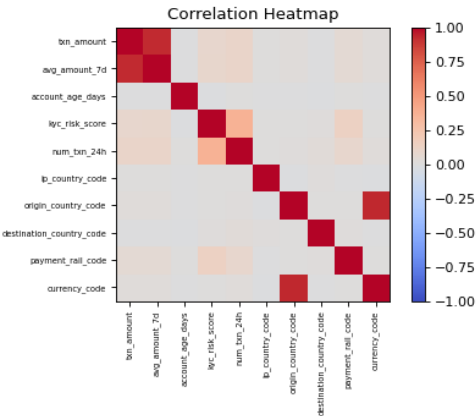
Charts



Feature Vs Feature

- No highly correlated feature pairs detected.

Charts



Time Drift

- Time series and drift analysis completed.

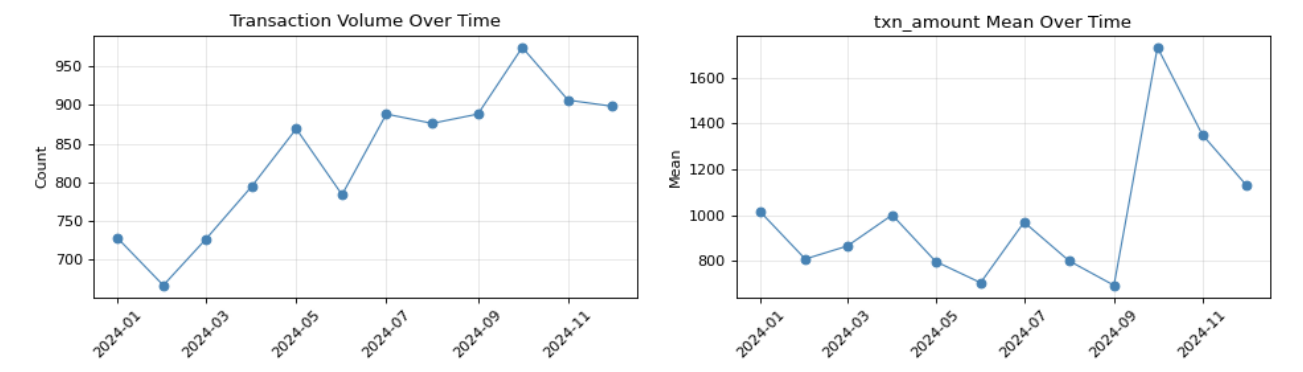
PSI Drift (Numeric)

Column	PSI
num_txn_24h	0.294
payment_rail_code	0.269
destination_country_code	0.185
kyc_risk_score	0.102
txn_amount	0.039
avg_amount_7d	0.029
ip_country_code	0.003
origin_country_code	0.002
currency_code	0.002
account_age_days	0.001

Categorical Drift (Total Variation)

Column	Drift Score
txn_type	0.292
payment_rail	0.238
destination_country	0.170
merchant_category	0.029
is_pep	0.027
txn_channel	0.023
currency	0.019
ip_country	0.016
origin_country	0.015
device_type	0.012

Charts



Skipped EDA Sections

No sections were skipped.

Run Configuration

Key	Value
data_path	bank_aml_tm_synthetic_10k_local=data/bank_aml_tm_synthetic_10k_local.csv
rows_original	10,000
rows_used	10,000
target_col	sar_label
time_col	value_date
time_parse_ratio	1
composition	{'mode': 'single_table', 'base_table': 'bank_aml_tm_synthetic_10k_local', 'tables': ['bank_aml_tm_synthetic_10k_local']}